

Milestone 3: Visualization & Exploratory Analysis

E-commerce Customer Behavior Analytics System

1. Executive Summary

Following the successful data standardization in Milestone 2, this phase transitions into **Exploratory Data Analysis (EDA)**. The objective is to visually decode relationships between demographics, loyalty tiers, and purchasing behaviors to support data-driven decision-making.

2. Visualization Strategy & Matrix

The system utilizes a multi-dimensional approach to ensure data perception is accurate and actionable.

Strategy	Metric Analyzed	Design Justification
Distribution	Total Spend	Histograms with KDE to identify spending density and consumer behavior patterns.
Proportion	Membership Tiers	Pie charts for a high-level view of customer loyalty across Gold, Silver, and Bronze.
Comparison	Satisfaction Levels	Bar charts for volume-based comparison of customer sentiment categories.
Correlation	Spend vs. Rating	Scatter plots (color-coded by Gender) to reveal experience vs. purchasing volume.

3. Advanced Exploratory Insights

- **Tier Impact:** Gold members exhibit the highest spending variance, suggesting that loyalty benefits drive significant engagement.
- **Sentiment Drivers:** A strong correlation exists between 'Average Rating' and 'Items Purchased', defining the threshold of customer engagement.

4. Design & Perception Principles

To meet professional standards, the following principles were applied:

- **Minimalism:** Focused on data trends by removing non-essential visual clutter.
- **Accessibility:** High-contrast color palettes (Viridis/Skyblue) for readability.
- **Contextual Labeling:** Self-explanatory titles and dynamic axis labeling.

5. Technical Implementation Note

Visualizations were rendered using Seaborn and Matplotlib, leveraging the **Standardized Data** from the Milestone 2 pipeline to ensure statistical comparability.